CHRISTINE YE SHU BLACKSHAW

Princeton University | Civil and Environmental Engineering | Princeton, NJ

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RESEARCH INTERESTS

Tropical cyclone and multi-hazard risk; climate change impacts on extreme wind and coastal flood hazards; compound urban hazards; hazard forecasting, uncertainty quantification, and risk communication; infrastructure and community resilience.

EDUCATION

PRINCETON UNIVERSITY, Princeton, NJ **2021-2027**
Ph.D. Candidate, Civil and Environmental Engineering (anticipated)

- Advisor: Professor Ning Lin, Hurricane Hazards and Risk Analysis Group
- M.A. awarded May 2023 following completion of the qualifying examination

COLUMBIA UNIVERSITY, New York, NY **2017-2021**
B.S., Civil Engineering; Minor in Environmental Engineering

RESEARCH EXPERIENCE

PRINCETON UNIVERSITY, Princeton, NJ **2021-present**
Ph.D. Candidate / Researcher, Hurricane Hazards and Risk Analysis Group

- Real-time TC Multi-Hazard Forecasting (lead): integrated official tropical cyclone forecasts with wind, rainfall, storm-tide, and power outage models for real-time multi-hazard assessment, evaluated uncertainty propagation across meteorological forecasts and hazard modeling, and built an interactive platform enabling users to track evolving hazard estimates across forecast updates.
- Structural Design Winds Under Future Climates (lead): evaluated hurricane wind hazards and structural design codes under CMIP6 climate-model ensembles, finding substantial variability in projected wind hazards and structural implications; developed an interactive calculator enabling users to compare design wind estimates by location, building characteristics, and climate futures.
- Compound TC Flood and Extreme Heat Modeling in Urban Environments (lead): coupled urban-canopy model with atmospheric and inundation models to characterize compound tropical-cyclone flooding and winds under extreme heat in urban environments; enabled city-block-scale assessment of post-storm heat stress and hurricane–heat hazard interactions.
- Generated tropical cyclone and wind-hazard inputs under current and future climates for New York City energy-resilience modeling, supporting assessment of power-outage risk and renewable-energy integration.
- Past research includes integrating paleo-hurricane reconstruction with hydrodynamic modeling, historical reconstruction of urban storm and meteorological events in New Jersey, and investigating methods for wind driven synthetic hurricane track modeling.

COLUMBIA UNIVERSITY, New York, NY **2020-2021**
Student Researcher, Department of Civil Engineering and Engineering Mechanics

- Developed a subway flood-propagation and damage model to evaluate proposed flood protection in New York City against current and future hurricane flooding.
- Constructed a three-dimensional leaf-area-density model for Manhattan combining LiDAR, tree-census data, and USDA street-tree growth projections.

TEACHING AND MENTORING EXPERIENCE

PRINCETON UNIVERSITY
Assistant in Instruction, CEE 460: Risk Analysis **Spring 2024**
Assistant in Instruction, CEE 306: Hydrology: Water and Climate **Spring 2023, 2024**
Undergraduate and High School Research Mentor (10 students) **Summers 2022-2025**

POLYGENCE *Research Mentor* **2023-present**

AWARDS AND HONORS

SEAS Award for Excellence **2024**
NSF Graduate Research Fellowship **2022-2026**
Tullio J. Borri '51 Award in Civil Engineering **2021**

PUBLICATIONS AND MANUSCRIPTS

1. **Blackshaw, C. Y.**, & Lin, N. (2026). Climate Sensitivity of Tropical Cyclone Design Winds in the United States. Earth's Future. (Under review)
2. **Blackshaw, C. Y.**, Spaulding, A. C., & Lin, N. (2026). Real-Time Tropical Cyclone Multi-Hazard Forecasting. Geophysical Research Letters. (In preparation)
3. Xu, L., Lin, N., Perera, A. D., Spaulding, A. C., Wang, Z., Oppenheimer, M., **Blackshaw, C. Y.**, & Poor, H. V. (2026). Climate change exacerbates disparities of energy resilience in New York City. Nature Communications.
4. **Blackshaw, C. Y.**, Lin, N., Begmohammadi, A., & Bou-Zeid, E. (2025). Investigating the interaction of tropical cyclone-heatwave compound hazards in urban environments. Urban Climate, 64, 102667.
5. Miura, Y., **Blackshaw, C. Y.**, Zhang, M. S., Mandli, K. T., & Deodatis, G. (2025). Coastal storm-induced flooding risk of the New York City subway amid climate change. Transportation Research Part D: Transport and Environment, 149, 104974.
6. Begmohammadi, A., Lin, N., Xi, D., & **Blackshaw, C.** (2025). Assessing future coastal flood hazards from tropical cyclones in the Northeastern United States. Earth's Future, 13(11), e2025EF006063.
7. Begmohammadi, A., **Blackshaw, C. Y.**, Lin, N., Gori, A., Wallace, E., Emanuel, K., & Donnelly, J. P. (2024). Integrating Climatological-Hydrodynamic Modeling and Paleohurricane Records to Assess Storm Surge Risk. Journal of Geophysical Research: Oceans, 129(1), e2023JC020354.
8. Princeton University. (2022). Expanding Climate Services to Respond to Adaptation Challenges: Insights from Managed Retreat Practice. Policy Workshop Report. Princeton School of Public & International Affairs.

PRESENTATIONS

1. **Blackshaw, C.**, & Lin, N. (2025). Designing for Tropical Cyclone Wind Loads under Non-Stationary Climates. AGU Fall Meeting 2025.
2. **Blackshaw, C.**, Begmohammadi, A., Lin, N., & Bou-Zeid, E. (2024). Investigating the interaction of tropical cyclone-heatwave compound hazards in urban environments. AGU Fall Meeting 2024.
3. **Blackshaw, C.** (2024). Paleo-hurricane Data for Long-term Hurricane Risk Assessment. Invited panel presentation, WTW Natural Catastrophe Program.
4. **Blackshaw, C.**, Begmohammadi, A., Lin, N., & Bou-Zeid, E. (2024). Investigating the Interaction of Tropical Cyclone-Heatwave Compound Hazards in Urban Canopy Models. Columbia University Extreme Heat Workshop.
5. **Blackshaw, C.**, Begmohammadi, A., & Lin, N. (2023). Integrating Climatological-Hydrodynamic Modeling and Paleohurricane Records to Assess Storm Surge Risk. AGU Fall Meeting 2023.
6. **Blackshaw, C.**, Begmohammadi, A., & Lin, N. (2023). Integrating ADCIRC modeling and paleohurricane records to assess long-term storm surge risk. ADCIRC Users Group Meeting 2023.

ACADEMIC AND COMMUNITY ENGAGEMENT

PRINCETON ENVIRONMENT AND CLIMATE SCHOLAR **2024-2026**

- Research project lead (2025-2026): lead 24 PhD students in developing interdisciplinary research to evaluate cooling and energy demands across Princeton campus and situate campus actions within the broader energy-policy landscape.
- Faculty seminar coordinator (2024-2025): organize interdisciplinary climate seminars with faculty and students.

PRINCETON CIVIL AND ENVIRONMENTAL ENGINEERING GRAD COUNCIL *Member* **2023-present**

RELEVANT WORKSHOPS AND COURSEWORK

- SPI 591G: Climate Adaptation and the Coasts: Managed Transition (Fall 2021)
- Investing in Futures Graduate Cohort – Understanding Climate Perception Through Art (Fall 2022)
- SPI 586E: Topics in STEP - Natural Hazards and Disaster Policy (Fall 2024)
- AAAS CASE Workshop (Spring 2025)
- Munich Re Externship (Spring 2026)

TECHNICAL EXPERTISE

Programming & computing: Python, R, MATLAB, HPC workflows
Geospatial & visualization: QGIS, ArcGIS, ParaView
Hazard & climate models: PepC (TC model), ADCIRC, SFINCS, HEC-RAS, WRF, PUCM (urban canopy model), tropical cyclone rainfall and wind modeling, global climate models
Research software & web: Interactive scientific web platforms, hazard visualization and decision-support tools, HTML, AWS

ACTIVITIES AND INTERESTS

Company Member, The Abdul Latif Collective (TALC)	**2021-present**
Princeton Dance Festival; Princeton University Ballet	**2021-present**
Columbia/Barnard Dance Department Performances; Columbia Ballet Collaborative; Columbia Repertory Ballet; Columbia University Ballet Ensemble	**2018-2021**